plastids can be classified into different types:

Protoplastids, Amyloplastids, Leucoplastids, Etioplasts, Chloro-amyloplasts and Chromoplasts.

- Protoplasts contain brown carotenoids, chlorophyll a and chlorophyll c pigments
- Amyloplasts synthesizes starch and stores them as granules in the stroma. Some types of plastids contain enzymes for the synthesis of certain small compounds.
- The leucoplasts are the colourless plastids of varied shapes and sizes.
- Rhodoplasts contain chlorophyll a and chlorophyll d along with phycobilin and phycoerythrin pigments.
- Chloroplasts-occur in green plants are characterised by the presence of Chlorophyll a and Chlorophyll b.
- Chromoplasts synthesize and store pigments called carotenoids, which are red, orange, or yellow molecules that give some flowers and fruits their colour.

1.4.12.Chloroplasts

Chloroplasts are members of a group of plant organelles collectively known as plastids. These are associated with photosynthesis. Majority of the chloroplasts of the green plants are found in the mesophyll cells of the leaves. These are lens-shaped, oval, spherical, discoid or even ribbon-like organelles having variable length (5-10 μm) and width (2-4 μm). Their number varies from 1 per cell of the *Chlamydomonas*, a green alga to 20-40 per cell in the mesophyll. Of the two, the inner chloroplast membrane is relatively less permeable. The space limited by the inner membrane of the chloroplast is called the stroma. A number of organised flattened membranous sacs called the thylakoids, are present in the stroma. Thylakoids are arranged in stacks like the piles of coins called grana (singular: granum) or the intergranal thylakoids. In addition, there are flat membranous tubules called the stroma lamellae connecting the thylakoids of the different grana. The membrane of the thylakoids enclose a space called a lumen. The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins. It also contains small, double-stranded circular DNA molecules and ribosomes. Chlorophyll pigments are present in the thylakoids.

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